Weekly Business Review: Olist Marketplace

Take-Home Assignment | Analysis of the Olist Brazilian E-Commerce dataset (historical 2016-2018 data)

Executive overview

The pipeline processed 99,441 orders and built an order-level mart with one row per order. Of these, 96,478 orders are marked delivered (97.02%). Delivered order value, defined here as item price plus freight for delivered orders, totals **R$ 15,419,773.75**. This is an operational marketplace value metric, not audited accounting revenue.

TOTAL ORDERS	DELIVERED RATE	DELIVERED ORDER VALUE	LOW REVIEW RATE
99,441	97.02%	R$ 15,419,773.75	14.64%

North Star and supporting metrics

Metric	Formula	Observed value	Why it matters
North Star: delivered order value	Sum(item price + freight) for delivered orders	R$ 15,419,773.75	Measures value attached to completed deliveries; not accounting revenue.
Delivered order rate	Delivered orders / all orders	97.02%	Shows completion of the order lifecycle.
Low-review order rate	Orders with average review score <= 2 / orders with a review	14.64%	Guardrail for customer dissatisfaction.
On-time delivery rate	Delivered orders on/before estimate / delivered orders with both dates	91.89%	Tracks delivery reliability; excludes records missing dates.
Repeat-customer rate	Unique customers with 2+ orders / unique customers with 1+ orders	3.12%	Indicates whether customers return to purchase again; uses customer_unique_id.

Pipeline and data quality

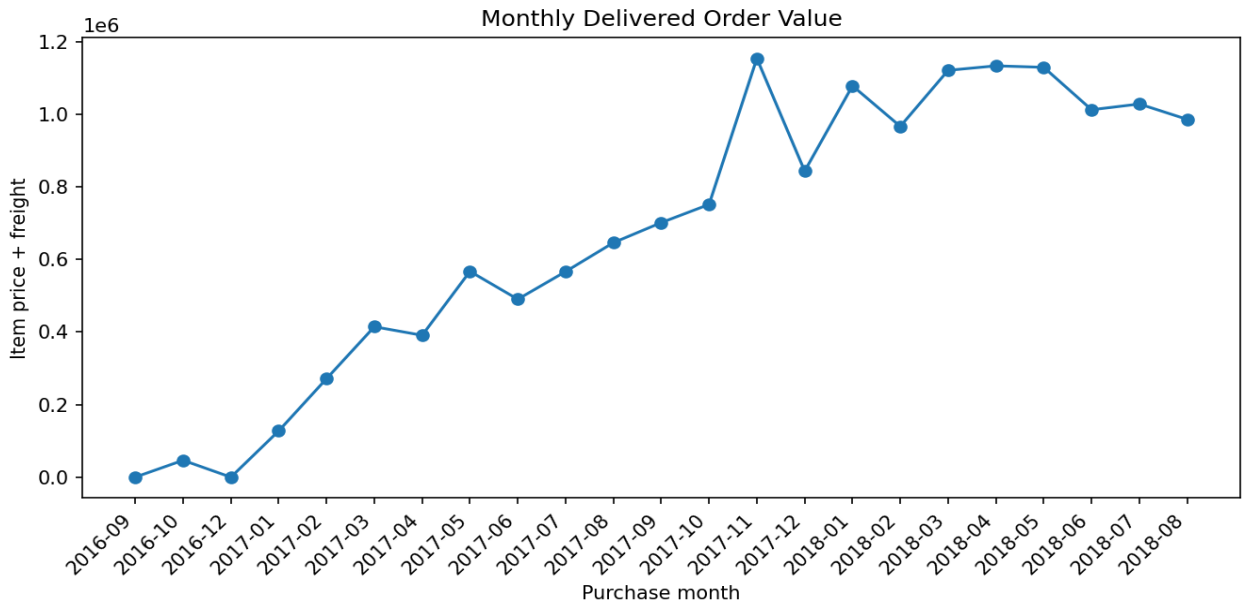
The pipeline loads raw tables, creates cleaned staging tables, builds the mart, and then evaluates SQL data-quality checks. The strict run stops before analytics if any check fails. A separate diagnostic run was used to generate charts and SQL outputs while retaining exceptions unchanged.

Check	Failing rows	Handling
Null/duplicate order ID	0	Pass; no action
Delivered before purchase	0	Pass; no action
Payment total vs. item + freight	260	Flagged for reconciliation; no source rows changed
Orders without customer	0	Pass; no action
Invalid review score	0	Pass; no action
Delivered order missing delivery timestamp	8	Flagged; delivery date remains unknown
Mart one-row-per-order consistency	0	Pass; no action

Data-quality counts are based on the pipeline checks. The 260 payment mismatches compare summed payment_value against summed item price + freight_value with a R$0.05 tolerance. These records require investigation and must not be silently dropped.

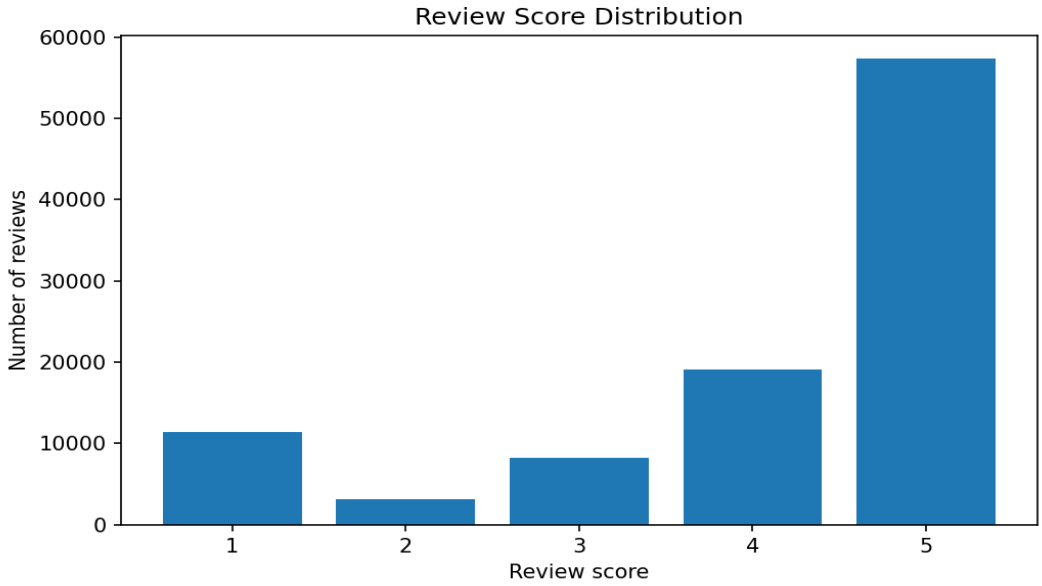
Performance trends and customer feedback

Chart 1. Monthly delivered order value



Delivered order value is based on item price plus freight grouped by purchase month. The earliest and latest months may be partial periods, so extreme growth percentages at the beginning of the series should not be interpreted as steady-state growth.

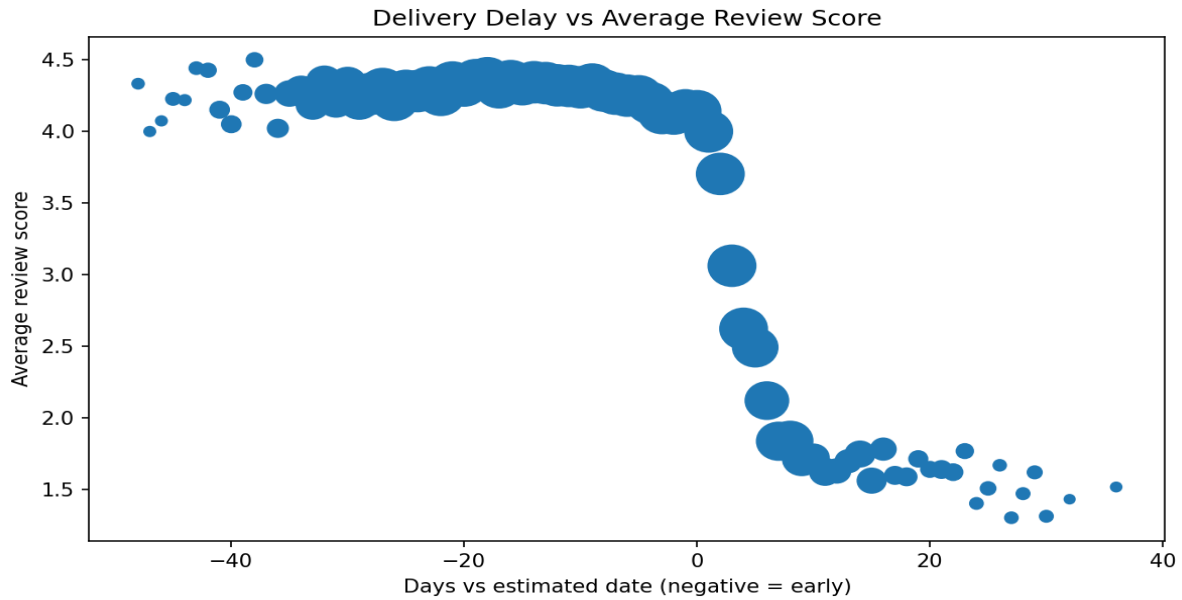
Chart 2. Review score distribution



The raw review table contains 99,224 review records; 14,575 records have scores 1-2 (14.69%). The order-level low-review guardrail is 14.64% because multiple reviews are averaged per order in the mart. These measures use different grains, so they differ slightly.

Delivery reliability and order activity

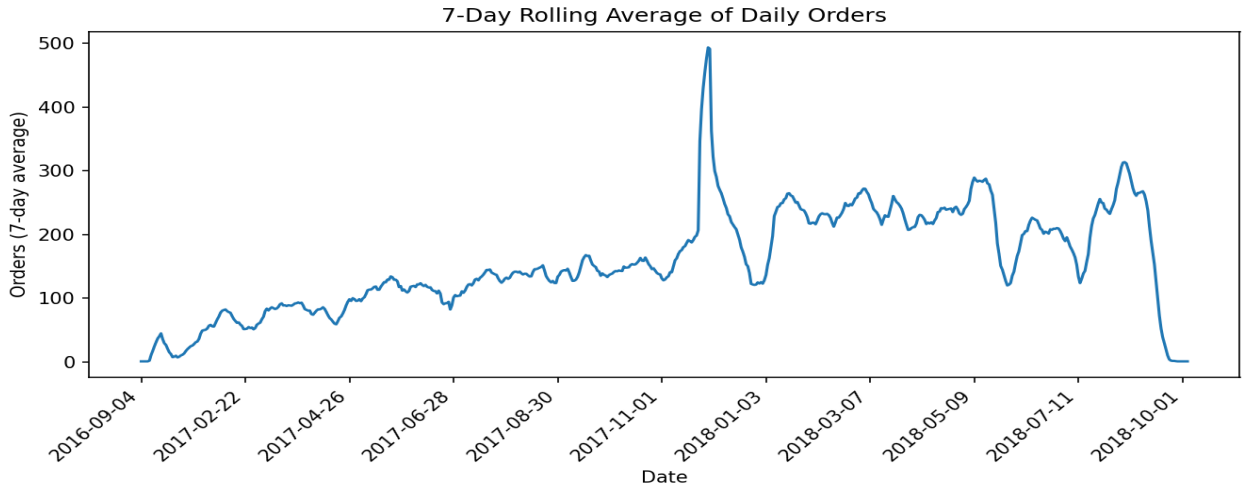
Chart 3. Delivery delay versus average review score



Delivery timing	Reviewed delivered orders	Average delay (days)	Average score	Low review rate
Early	88,163	-13.01	4.29	9.19%
Late	7,661	9.45	2.57	53.99%

Among reviewed delivered orders with delay data, late orders have an average review score of 2.57 and a 53.99% low-review rate; early orders average 4.29 with a 9.19% low-review rate. This is a strong association, not proof that delivery delay alone causes a low score.

Chart 4. Seven-day rolling average of daily order count



The SQL query uses a calendar date spine and includes zero-order days, so the rolling average represents seven calendar days rather than seven dates with activity.

SQL analytics results and interpretation

All five analytics questions are answered with separate SQL files using CTEs and window functions where appropriate. Output CSV files are saved in `reports/`.

1. Month-over-month delivered order value

Month	Order value	Previous month	Growth
2018-03	R$ 1,120,598.24	R$ 966,168.41	15.98%
2018-04	R$ 1,132,878.93	R$ 1,120,598.24	1.10%
2018-05	R$ 1,128,774.52	R$ 1,132,878.93	-0.36%
2018-06	R$ 1,011,978.29	R$ 1,128,774.52	-10.35%
2018-07	R$ 1,027,807.28	R$ 1,011,978.29	1.56%
2018-08	R$ 985,491.64	R$ 1,027,807.28	-4.12%

2. Top three categories by customer state (example: Sao Paulo)

Rank	Category	Item revenue
1	bed_bath_table	R$ 478,284.52
2	health_beauty	R$ 462,305.22
3	watches_gifts	R$ 435,009.92

3. Seven-day rolling average: computed for all calendar dates; full day-by-day results are in `03_rolling_7day_order_count.csv`.

4. Repeat-purchase rate: 2,997 of 96,096 unique customers made at least two orders (3.12%). The query counts customer_unique_id rather than customer_id because customer_id can differ between orders for the same underlying customer.

5. Delivery delay versus review score: the grouped results are shown on page 3.

Highest delivered order value by customer state

State	Delivered orders	Delivered order value
SP	40,501	R$ 5,769,703.15
RJ	12,350	R$ 2,055,401.57
MG	11,354	R$ 1,818,891.67
RS	5,345	R$ 861,472.79
PR	4,923	R$ 781,708.80

SQL correctness note: order-level item and payment totals are aggregated before joining to the mart to avoid multiplying orders when an order has multiple items or payment rows. Geolocation is loaded raw but not joined into the mart because zip prefixes are not unique.

Memo to the Vice President

To: Vice President, Marketplace Operations | From: Data Analyst | Subject: Reduce late deliveries to improve customer satisfaction

Recommendation

Prioritize late-delivery reduction as a customer-satisfaction initiative. Start with a weekly carrier and region review for orders predicted to miss their estimated delivery date, and trigger proactive customer notifications with a revised ETA when risk is identified. Monitor the effect using the low-review rate and on-time delivery rate.

Evidence

The mart contains 99,441 orders, of which 96,478 are marked delivered. The delivered-order rate is 97.02%, and delivered order value is R$ 15,419,773.75. For reviewed delivered orders with available delivery dates, 7,661 were late and 88,163 were early. Late orders averaged 2.57/5, compared with 4.29/5 for early deliveries. The low-review rate (scores 1-2) was 53.99% for late deliveries versus 9.19% for early deliveries - a 44.80 percentage-point difference.

Proposed action

1. Flag at-risk shipments before the estimated delivery date using carrier scan events and historical delivery performance. 2. Contact customers proactively when a delay is likely and provide a revised ETA. 3. Review late shipments weekly by carrier, seller, and customer state; direct operational follow-up to the largest recurring delay sources. 4. Compare on-time rate and low-review rate before and after the intervention, while checking that cancellations and customer contacts do not rise.

Measurement and caveats

Track on-time delivery rate (currently 91.89% among delivered orders with both dates) and low-review rate (currently 14.64% across reviewed orders). The dataset is historical, covering primarily 2016-2018, so this establishes a historical relationship rather than current performance. The association does not prove causality; test the intervention and measure change. There are 260 payment mismatches and 8 delivered orders missing delivery timestamps, which remain flagged for data-quality follow-up.

Decision requested: approve a focused pilot for late-delivery risk alerts and weekly carrier exception review, with on-time delivery and low-review rate as the primary outcome measures.